

Precalculus Review (Part II)

- I) Simplifying Complex Fractions
- II) Logarithms
- III) Function Notation
- IV) Piecewise Functions
- V) Using the Library of Functions to Graph by Transformations

Simplifying Complex Fractions

- ① $\frac{\frac{1}{2}}{\frac{1}{4}} = \frac{1}{2} \div \frac{1}{4} = \frac{1}{2} \cdot \frac{4}{1} = \frac{4}{2} = \boxed{2}$
- ② $\frac{\frac{2}{3}}{\frac{5}{9}} = \frac{2}{3} \div \frac{5}{9} = \frac{2}{3} \cdot \frac{9}{5} = \frac{18}{15} = \boxed{\frac{6}{5}}$
- ③ $\frac{\frac{3}{10}}{\frac{4}{5}} = \frac{3}{10} \div \frac{4}{5} = \frac{3}{10} \cdot \frac{5}{4} = \frac{15}{40} = \boxed{\frac{3}{8}}$
- ④ $\frac{6}{\frac{2}{9}} = 6 \div \frac{2}{9} = \frac{6}{1} \div \frac{2}{9} = \frac{6}{1} \cdot \frac{9}{2} = \frac{54}{2} = \boxed{27}$

Logarithms

Logarithm Property I

Logarithmic equations and exponential equations are equivalent. In other words, $x = \log_b C$ if and only if $C = b^x$. We learned in geometry class that "if and only if" means the converse of the statement is true also. Note: Traditionally, neither "C" nor "b" is allowed to be negative or zero when placed in a logarithmic expression.

Logarithm Property II

$$\log_b b^x = x$$

Logarithm Property III

$$\log_b b = 1$$

Logarithm Property IV

$$b^{\log_b x} = x$$

Logarithm Property V

$$\log_k(AB) = \log_k A + \log_k B$$

Logarithm Property VI

$$\log_k\left(\frac{A}{B}\right) = \log_k A - \log_k B$$

Logarithm Property VII

$$\log_k A^B = B \log_k A$$

Logarithm Property VIII

$$\text{If } \log_k A = \log_k B, \text{ then } A = B.$$

Determine if the following values can be discovered without a calculator. Then find the values. If a calculator is necessary, round your answer to the nearest thousandth.

⑤ $\log_3 27$

$$x = \log_3 27$$

$$3^x = 27$$

$$3^x = 3^3$$

$$x = 3$$

No Calculator

$$x = \log_3 27 = \boxed{3}$$

⑥ $\log_8 15$

$$x = \log_8 15$$

$$8^x = 15$$

Calculator

$$\log_8 15 \approx \boxed{1.302}$$

⑦ $\log_2 16$

$$x = \log_2 16$$

$$2^x = 16$$

$$2^x = 2^4$$

$$x = 4$$

No Calculator

$$x = \log_2 16 = \boxed{4}$$

⑧ $\log 1000$

$$x = \log_{10} 1000$$

$$10^x = 1000$$

$$10^x = 10^3$$

$$x = 3$$

No Calculator

$$x = \log 1000 = \boxed{3}$$

⑨ $\ln 1$

$$x = \log_e 1$$

$$e^x = 1$$

$$e^x = e^0$$

$$x = 0$$

No Calculator

$$x = \ln 1 = \boxed{0}$$

⑩ $\ln e^4$

$$x = \log_e e^4$$

$$e^x = e^4$$

$$x = 4$$

No Calculator

$$x = \ln e^4 = \boxed{4}$$

⑪ $\ln 7$

$$x = \log_e 7$$

$$e^x = 7$$

Calculator

$$\ln 7 \approx \boxed{1.946}$$

⑫ $\ln e$

$$x = \log_e e$$

$$e^x = e$$

$$e^x = e^1$$

$$x = 1$$

No Calculator

$$x = \ln e = \boxed{1}$$

⑬ $\log_4 16$

$$x = \log_4 16$$

$$4^x = 16$$

$$4^x = 4^2$$

$$x = 2$$

No Calculator

$$x = \log_4 16 = \boxed{2}$$

⑭ $\log_3 28$

$x = \log_3 28$

$3^x = 28$

Calculator

$\log_3 28 \approx \boxed{3.033}$

⑮ $\log_5 125$

$x = \log_5 125$

$5^x = 125$

$5^x = 5^3$

$x = 3$

No calculator

$x = \log_5 125 = \boxed{3}$

⑯ $\log 5$

$x = \log_{10} 5$

$10^x = 5$

Calculator

$x = \log 5 \approx \boxed{0.699}$

⑰ $\log 10$

$x = \log_{10} 10$

$10^x = 10$

$10^x = 10^1$

$x = 1$

No calculator

$x = \log 10 = \boxed{1}$

⑱ $\ln e^6$

$x = \log_e e^6$

$e^x = e^6$

$x = 6$

No calculator

$x = \ln e^6 = \boxed{6}$

• ⑲ $\ln\left(\frac{1}{e^2}\right)$

$x = \log_e \left(\frac{1}{e^2}\right)$

$e^x = \frac{1}{e^2}$

$e^x = e^{-2}$

$x = -2$

No calculator

$x = \ln\left(\frac{1}{e^2}\right) = \boxed{-2}$

• ⑳ $\log_3 \sqrt[4]{3}$

$x = \log_3 3^{\frac{1}{4}}$

$3^x = 3^{\frac{1}{4}}$

$x = \frac{1}{4}$

No calculator

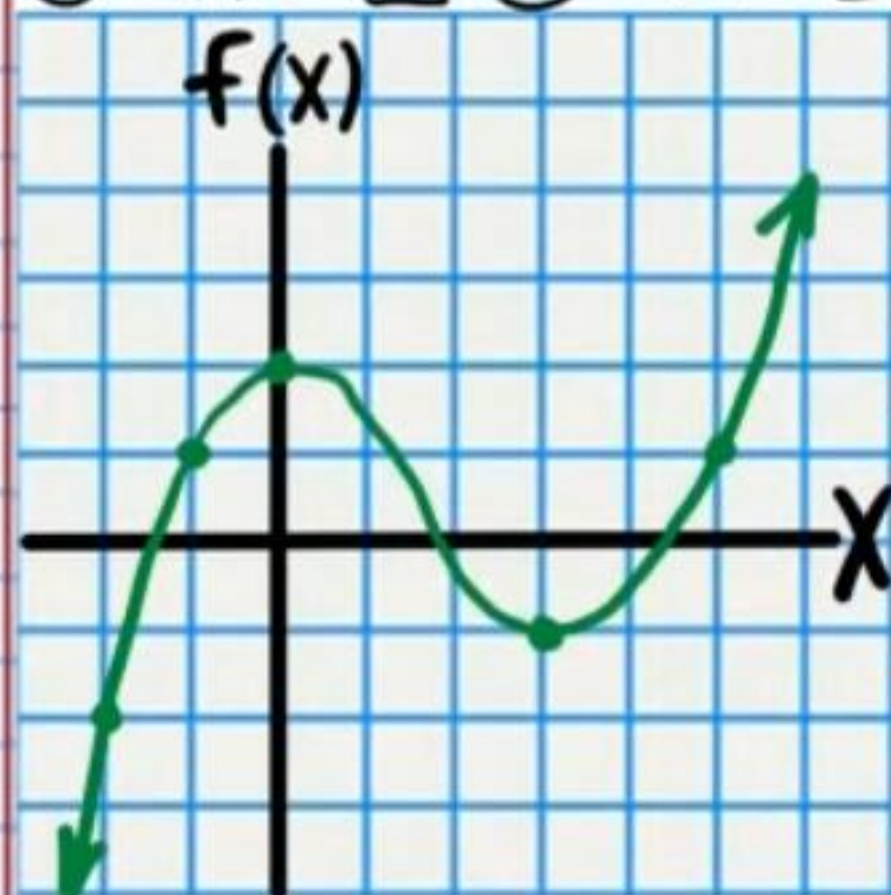
$x = \log_3 \sqrt[4]{3} = \boxed{\frac{1}{4}}$

Function Notation

Use the graphs below to find the following values.

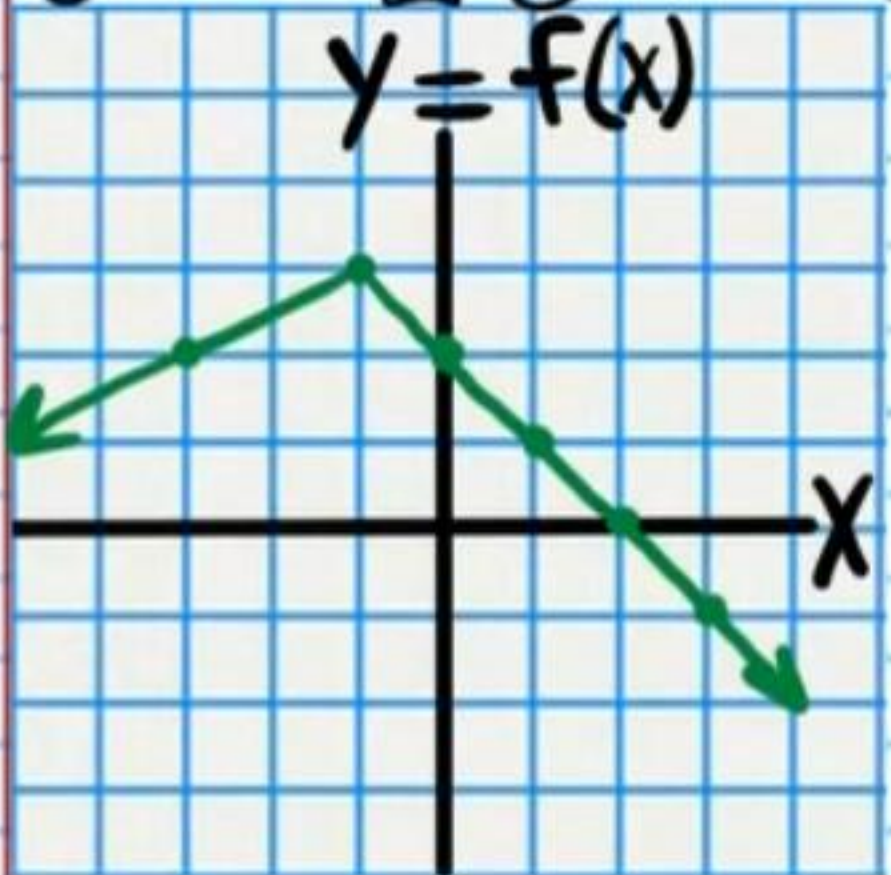
• (21) $f(3) = \boxed{-1}$ (22) $f(-1) = \boxed{1}$

(23) $f(0) = \boxed{2}$ (24) $f(5) = \boxed{1}$



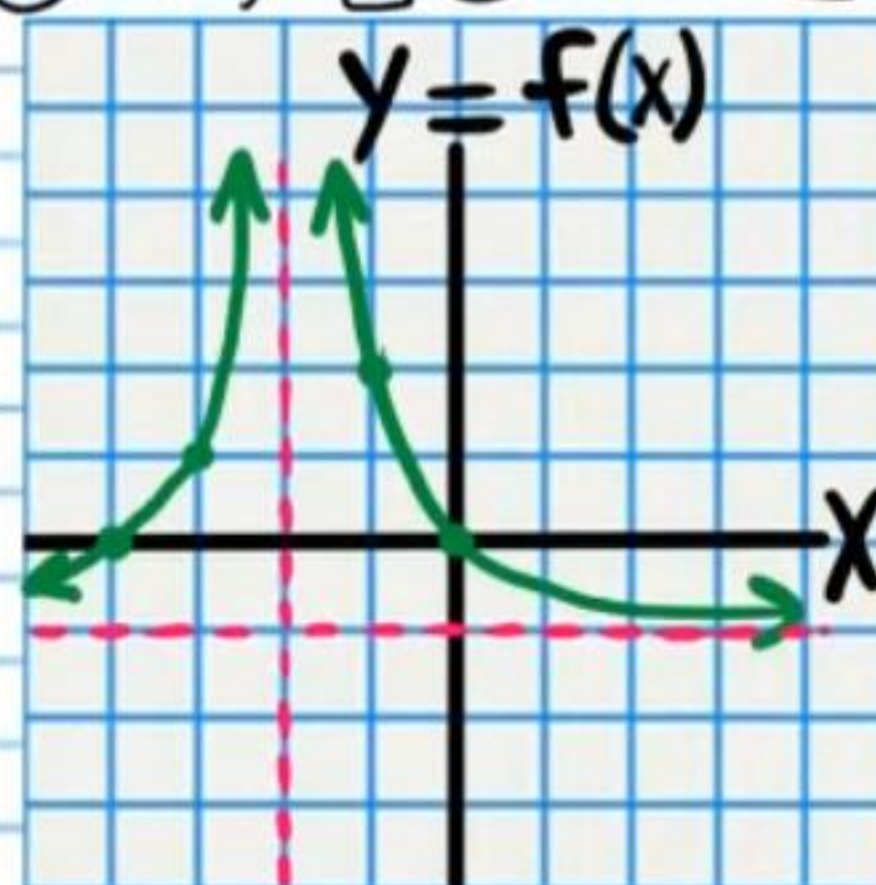
• (29) $f(2) = \boxed{0}$ (30) $f(1) = \boxed{1}$

(31) $f(-1) = \boxed{3}$ (32) $f(-3) = \boxed{2}$



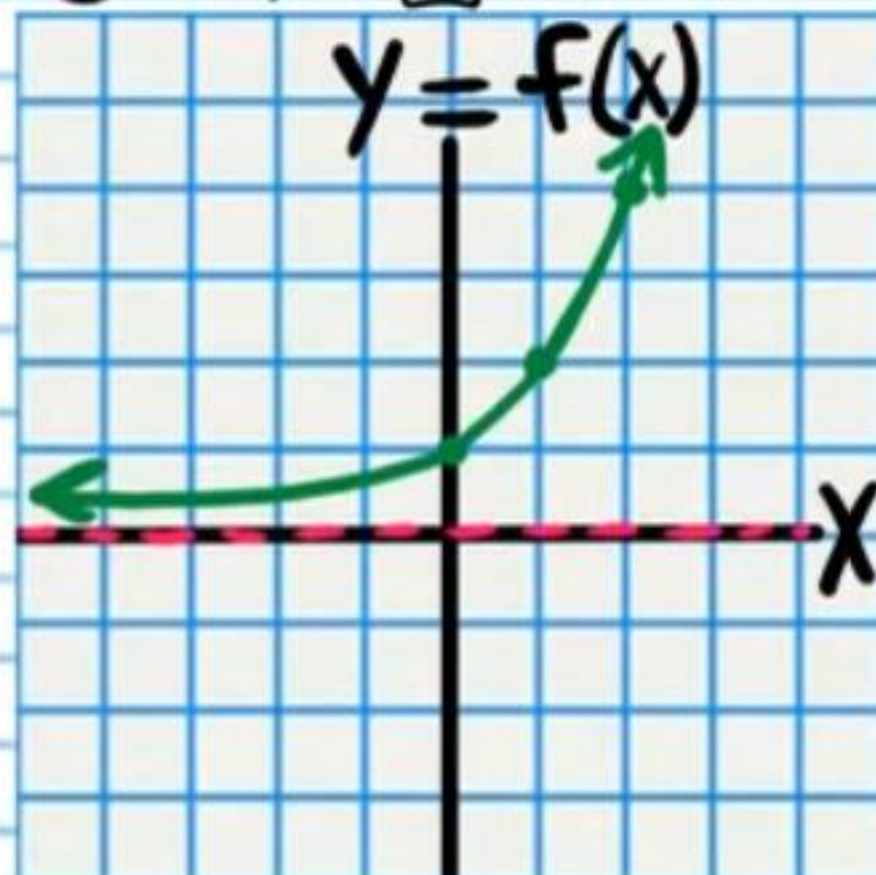
(25) $f(-3) = \boxed{1}$ (26) $f(0) = \boxed{0}$

(27) $f(-1) = \boxed{2}$ (28) $f(-4) = \boxed{0}$



(33) $f(0) = \boxed{1}$ (34) $f(1) = \boxed{2}$

(35) $f(2) = \boxed{4}$



Piecewise Functions

Use the piecewise functions to find the values of the expressions.

• (36) $f(x) = \begin{cases} x+8, & -10 < x \leq 4 \\ \ln x, & 4 < x \leq 22 \end{cases}$

a) $f(-6) = -6+8 = \boxed{2}$

b) $f(4.1) = \boxed{\ln(4.1)}$

c) $f(4) = 4+8 = \boxed{12}$

d) $f(e^2) = \ln e^2 = \log_e e^2 = \boxed{2}$

(37) $f(x) = \begin{cases} 5^{x+1} - 1, & x > 2 \\ (x+4)^3, & x \leq 2 \end{cases}$

a) $f(-4) = (-4+4)^3 = 0^3 = \boxed{0}$

b) $f(3) = 5^{3+1} - 1 = 5^4 - 1 = 625 - 1 = \boxed{624}$

c) $f(2) = (2+4)^3 = 6^3 = \boxed{216}$

d) $f(-2) = (-2+4)^3 = 2^3 = \boxed{8}$

$$\textcircled{38} f(x) = \begin{cases} 5, & -8 \geq x \\ |-x| + 4, & -8 < x \end{cases}$$

$$a) f(-7) = |-(-7)| + 4 = |7| + 4 = 7 + 4 = \boxed{11}$$

$$b) f(100) = |-100| + 4 = 100 + 4 = \boxed{104}$$

$$c) f(-8) = \boxed{5}$$

$$d) f(-9) = \boxed{5}$$

$$\bullet \textcircled{39} f(x) = \begin{cases} 16 - x, & x < 0 \\ \sqrt{x}, & x \geq 0 \end{cases}$$

$$a) f(100) = \sqrt{100} = \boxed{10}$$

$$b) f(-9) = 16 - (-9) = \boxed{25}$$

$$c) f(0) = \sqrt{0} = \boxed{0}$$

$$d) f(-8) = 16 - (-8) = 16 + 8 = \boxed{24}$$

$$\textcircled{40} f(x) = \begin{cases} \frac{1}{x+7}, & -7 < x \\ 1+2x, & -7 \geq x \end{cases}$$

$$a) f(-5) = \frac{1}{-5+7} = \boxed{\frac{1}{2}}$$

$$b) f(0) = \frac{1}{0+7} = \boxed{\frac{1}{7}}$$

$$c) f(-10) = 1 + 2(-10) = 1 - 20 = \boxed{-19}$$

$$d) f(-3\pi) = 1 + 2(-3\pi) = \boxed{1 - 6\pi}$$

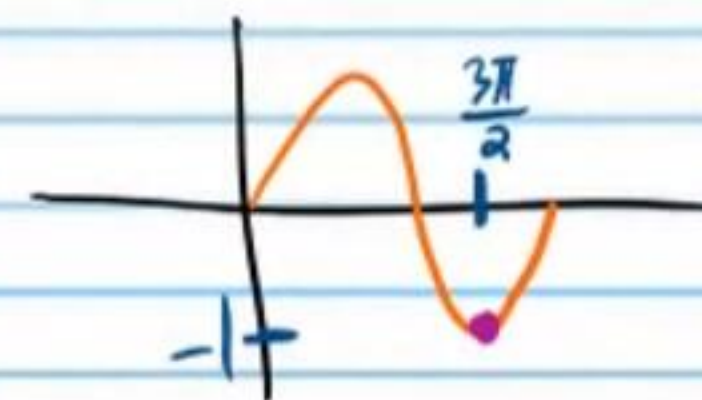
$$\textcircled{41} f(x) = \begin{cases} x, & x \leq \pi \\ \sin x, & x > \pi \end{cases}$$

$$a) f(2\pi) = \sin(2\pi) = \boxed{0}$$

$$b) f\left(\frac{\pi}{2}\right) = \boxed{\frac{\pi}{2}}$$

$$c) f\left(\frac{3\pi}{2}\right) = \sin\left(\frac{3\pi}{2}\right) = \boxed{-1}$$

$$d) f(\pi) = \boxed{\pi}$$



$$\textcircled{42} f(x) = \begin{cases} \log_6(x-1), & 1 < x \\ 800, & 1 \geq x \end{cases}$$

$$a) f(1) = \boxed{800}$$

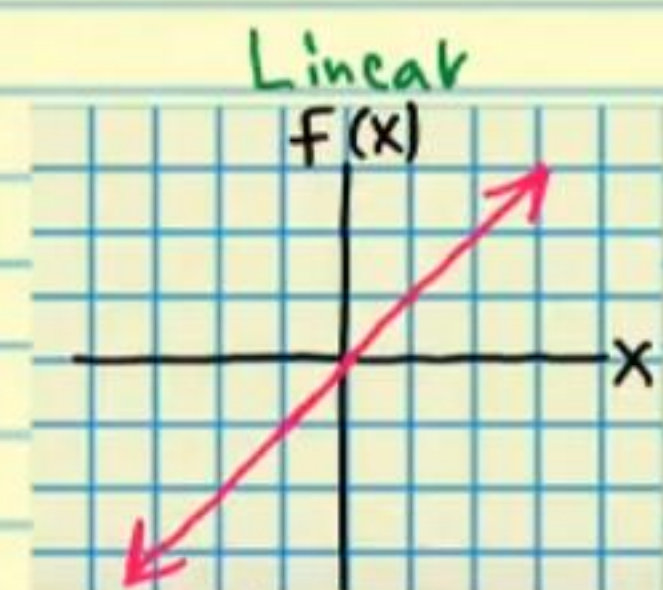
$$b) f(2) = \log_6(2-1) = \log_6 1 = \boxed{0}$$

$$c) f(0) = \boxed{800}$$

$$d) f(37) = \log_6(37-1) = \log_6 36 = \boxed{2}$$

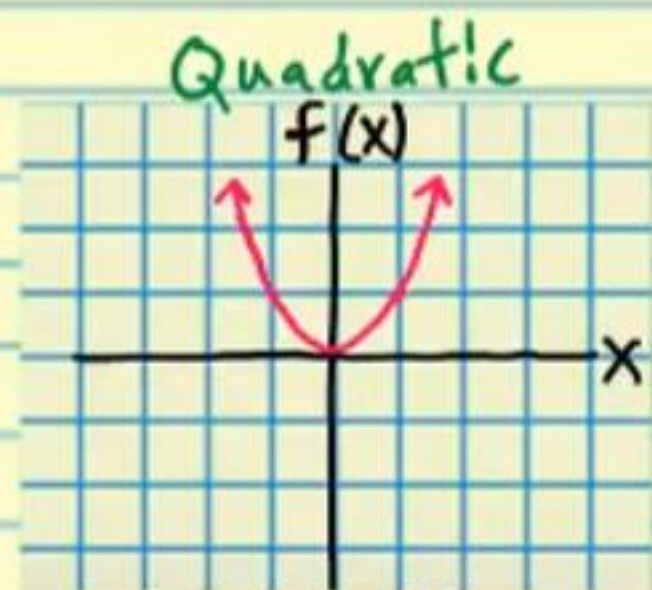
The Library of Functions

★ Also called "parent" functions.



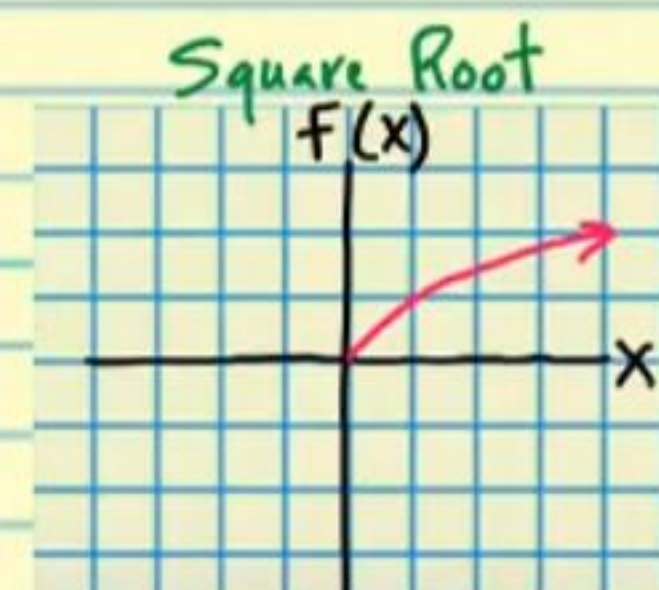
$$f(x) = x$$

x	-2	-1	0	1	2
f(x)	-2	-1	0	1	2



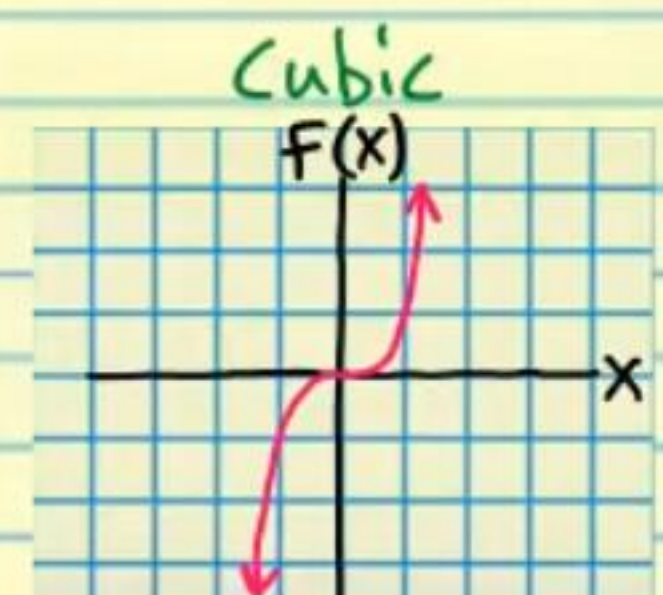
$$f(x) = x^2$$

x	-2	-1	0	1	2
f(x)	4	1	0	1	4



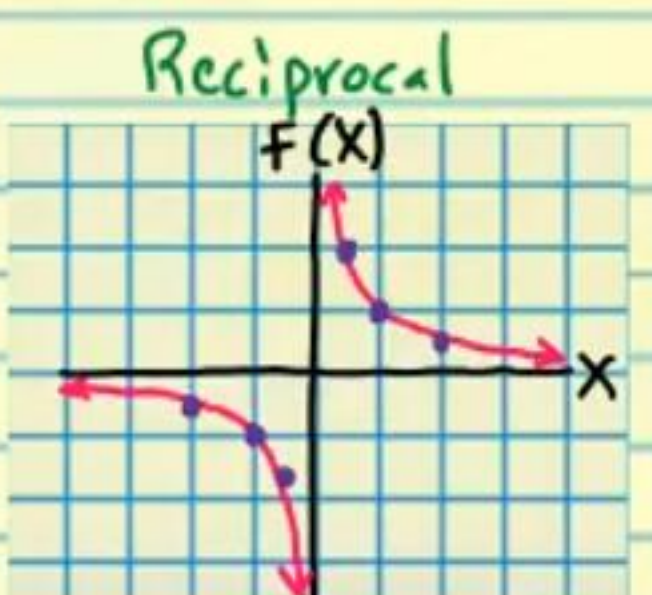
$$f(x) = \sqrt{x}$$

x	0	1	4	9	16
f(x)	0	1	2	3	4



$$f(x) = x^3$$

x	-2	-1	0	1	2
f(x)	-8	-1	0	1	8



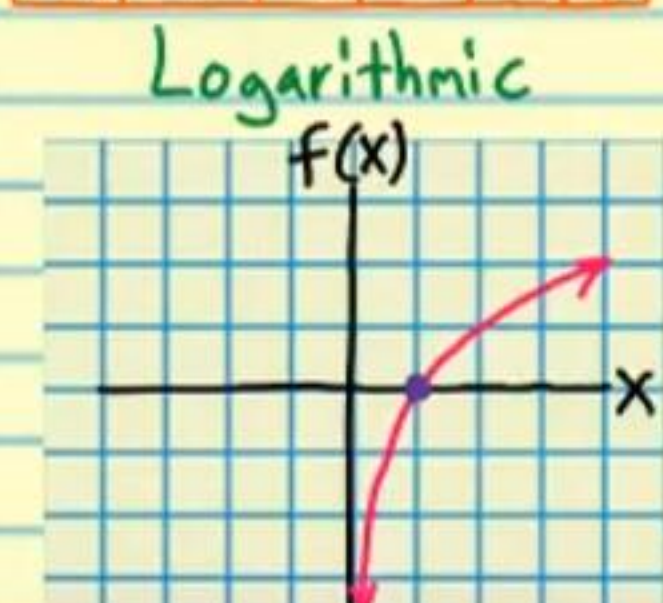
$$f(x) = \frac{1}{x}$$

x	-2	-1	-1/2	0	1/2	1	2
f(x)	-1/2	-1	-2	und	2	1	1/2



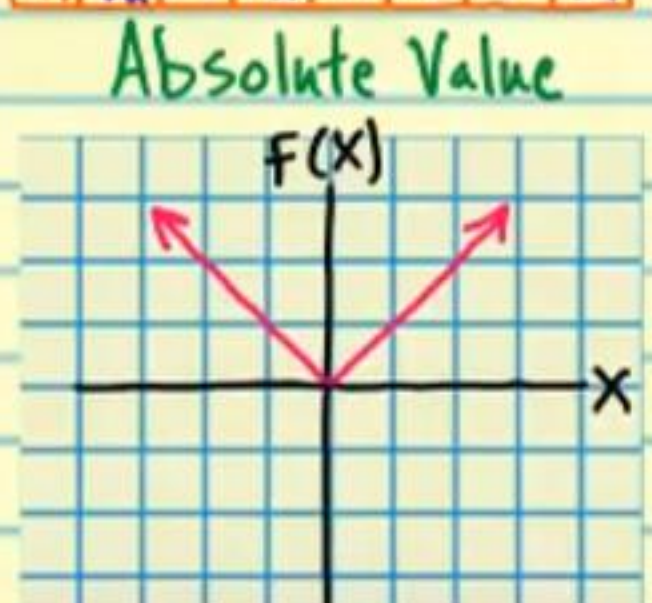
$$f(x) = a^x, a > 1$$

x	-	-	0	-	-
f(x)	-	-	1	-	-



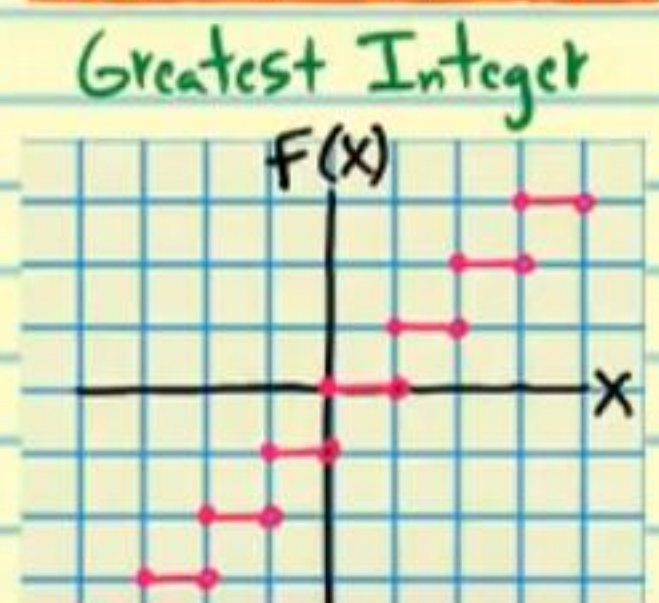
$$f(x) = \log_a x$$

x	0	1	-	-	-
f(x)	und	0	-	-	-



$$f(x) = |x|$$

x	-2	-1	0	1	2
f(x)	2	1	0	1	2



$$f(x) = [x]$$

x	-2 1/2	-1	0	1/2	1 1/2
f(x)	-3	-1	0	0	1

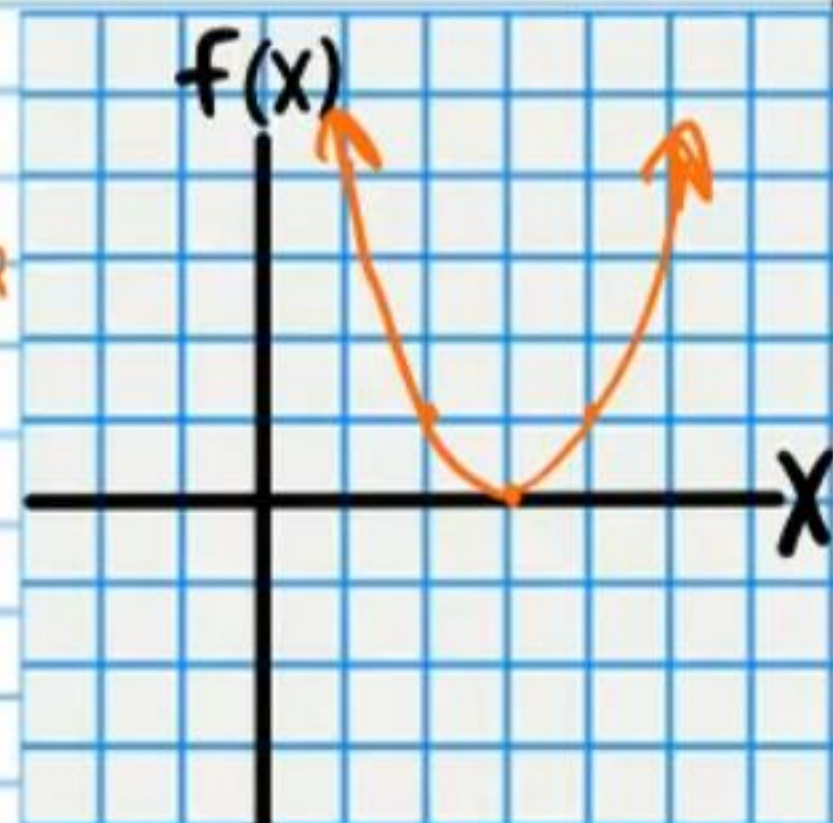
Graphing by Transformations

Graph the following functions.

• (43) $f(x) = (x-3)^2$

$$f(x) = x^2$$

$$f(x) = (x-3)^2$$



(44) $f(x) = \frac{1}{-x} + 2$

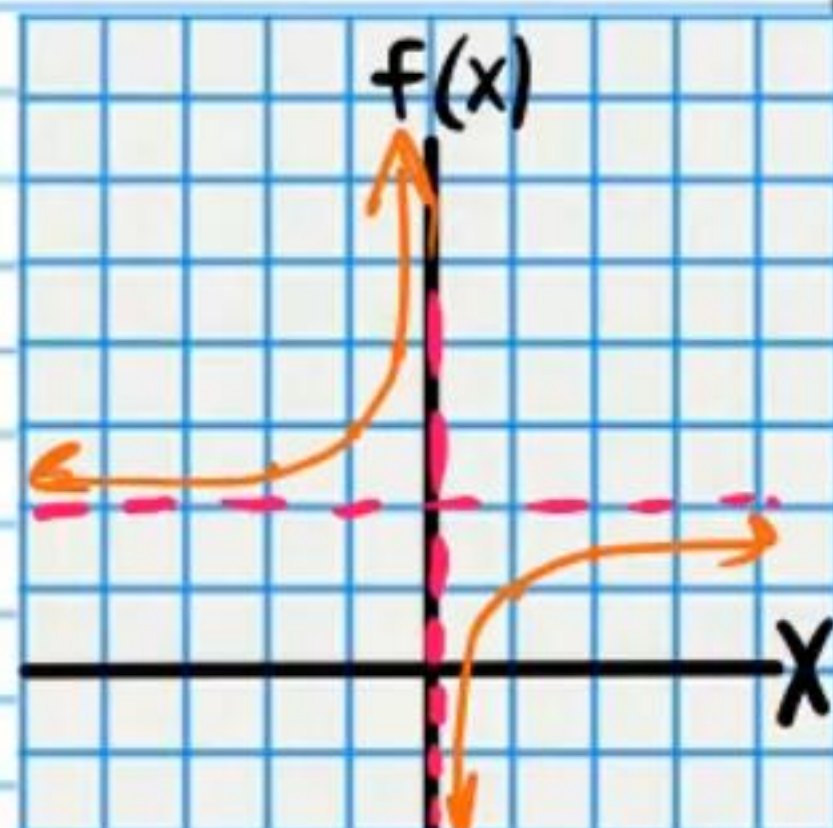
$$y = \frac{1}{-x} + 2$$

$$y - 2 = \frac{1}{-x}$$

$$y = \frac{1}{x}$$

$$y = \frac{1}{-x}$$

$$y - 2 = \frac{1}{-x}$$



(45) $f(x) = -\frac{1}{2}|x+1| - 1$

$$y = -\frac{1}{2}|x+1| - 1$$

$$y + 1 = -\frac{1}{2}|x+1|$$

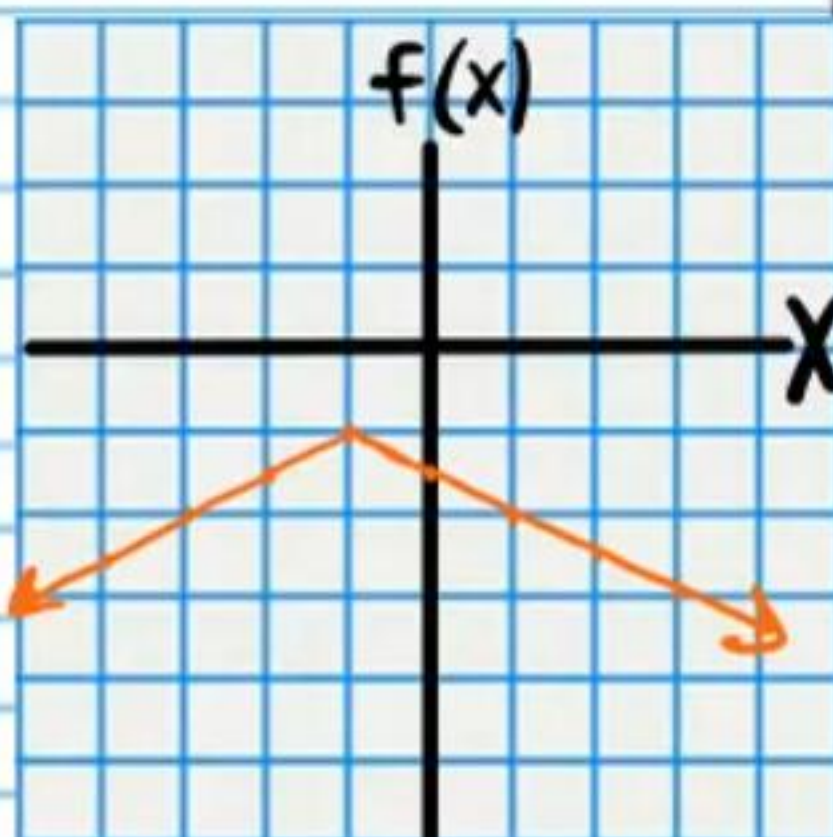
$$-2(y+1) = |x+1|$$

$$y = |x|$$

$$2y = |x|$$

$$-2y = |x|$$

$$-2(y+1) = |x+1|$$



④⑥ $f(x) = -\sqrt{x}$

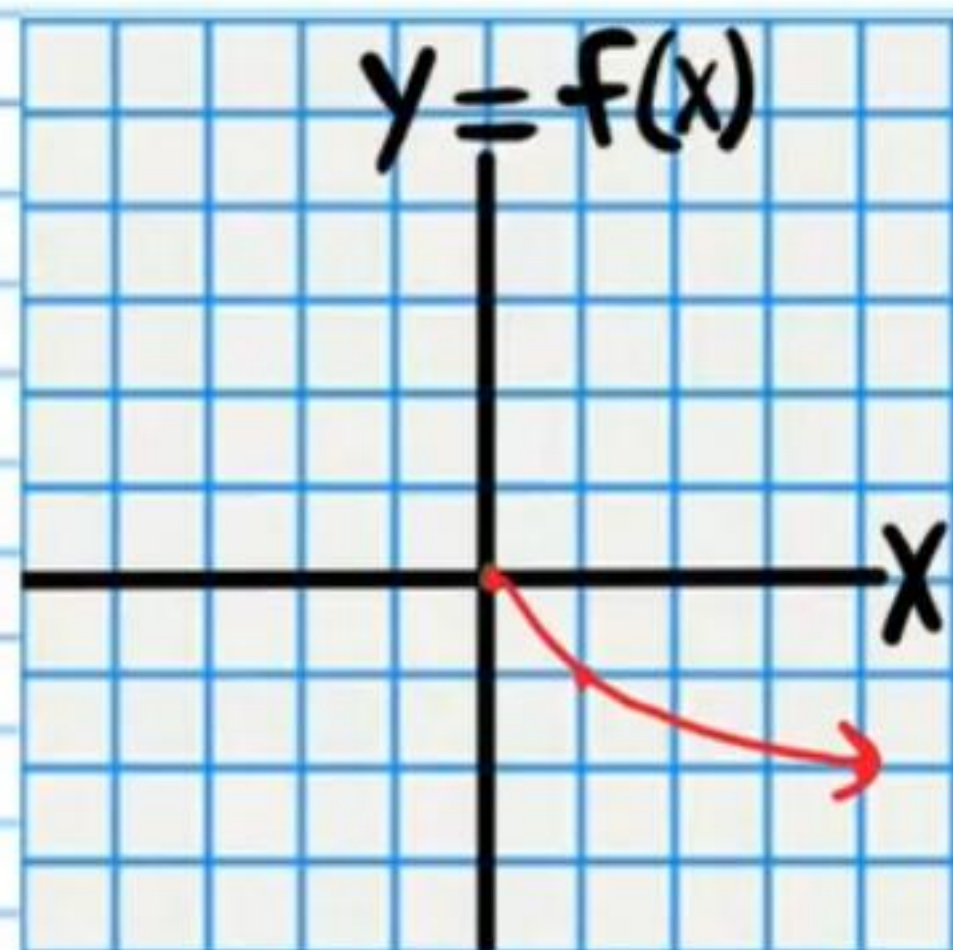
↓

$y = -\sqrt{x}$

$-y = \sqrt{x}$

$y = \sqrt{x}$

$-y = \sqrt{x}$



④⑦ $f(x) = 2(x+1)^3$

↓

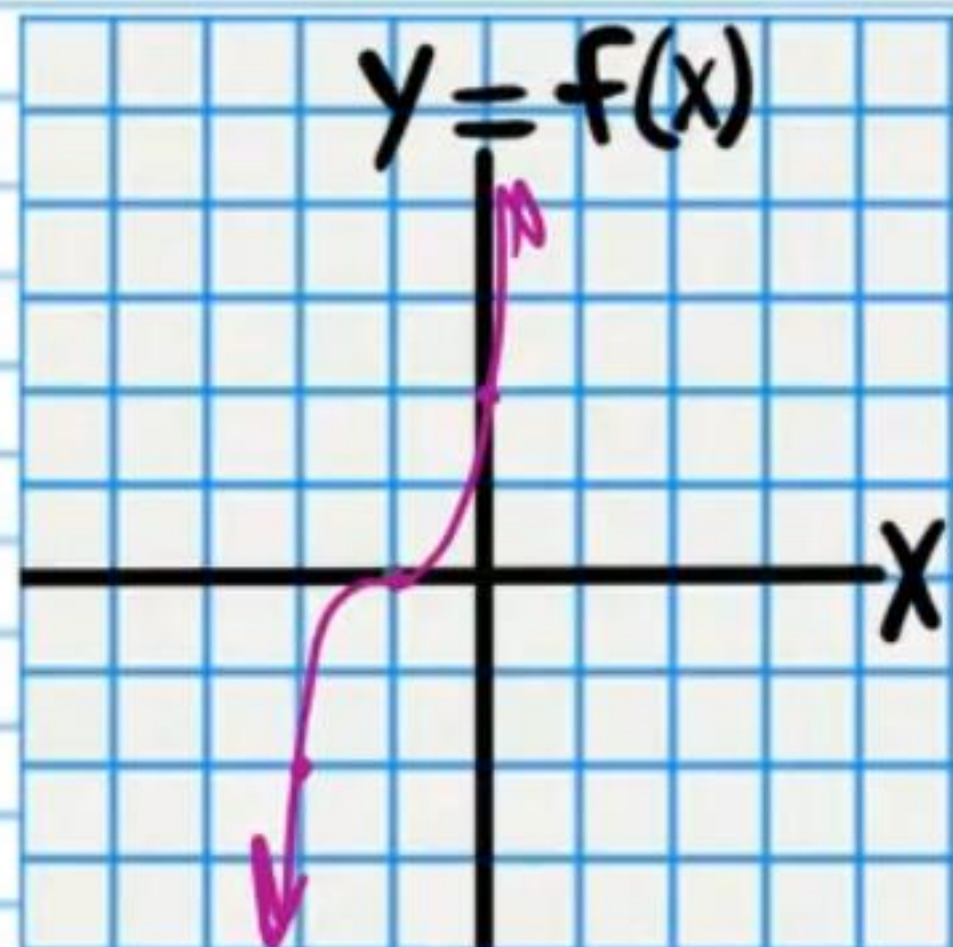
$y = 2(x+1)^3$

$\frac{y}{2} = (x+1)^3$

$y = x^3$

$y = (x+1)^3$

$\frac{y}{2} = (x+1)^3$

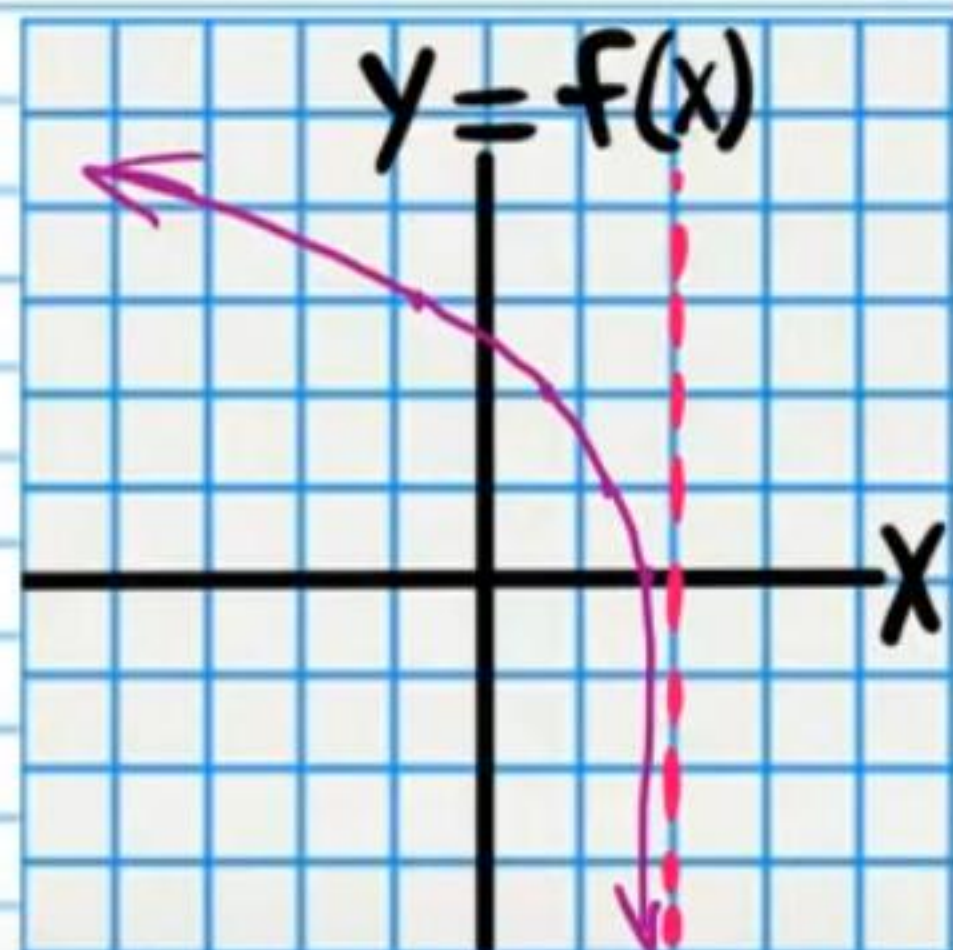


④⑧ $f(x) = \log_2(-3x+6)$

$y = \log_2 x$

$y = \log_2(x+6)$

$y = \log_2(-3x+6)$



• ④⑨ $f(x) = 2|x|$

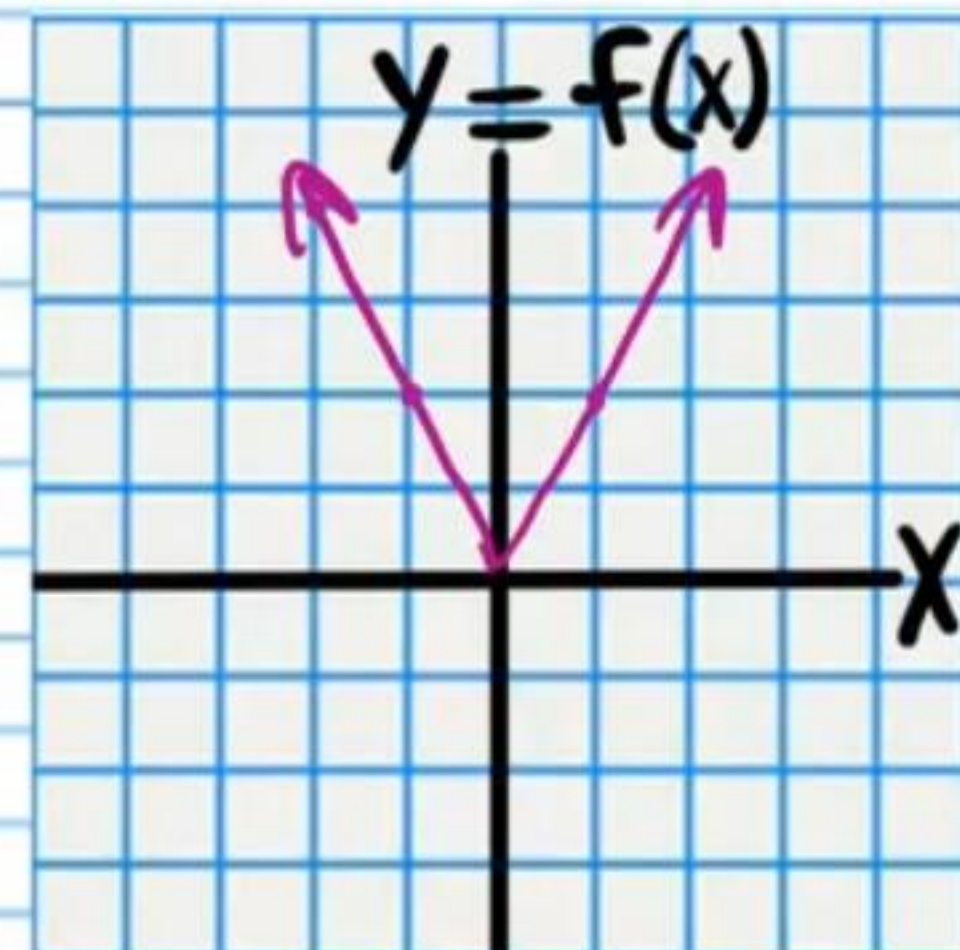
↓

$y = 2|x|$

$\frac{y}{2} = |x|$

$y = |x|$

$\frac{y}{2} = |x|$



⑤⑩ $f(x) = -\frac{x^2}{3}$

↓

$y = -\frac{x^2}{3}$

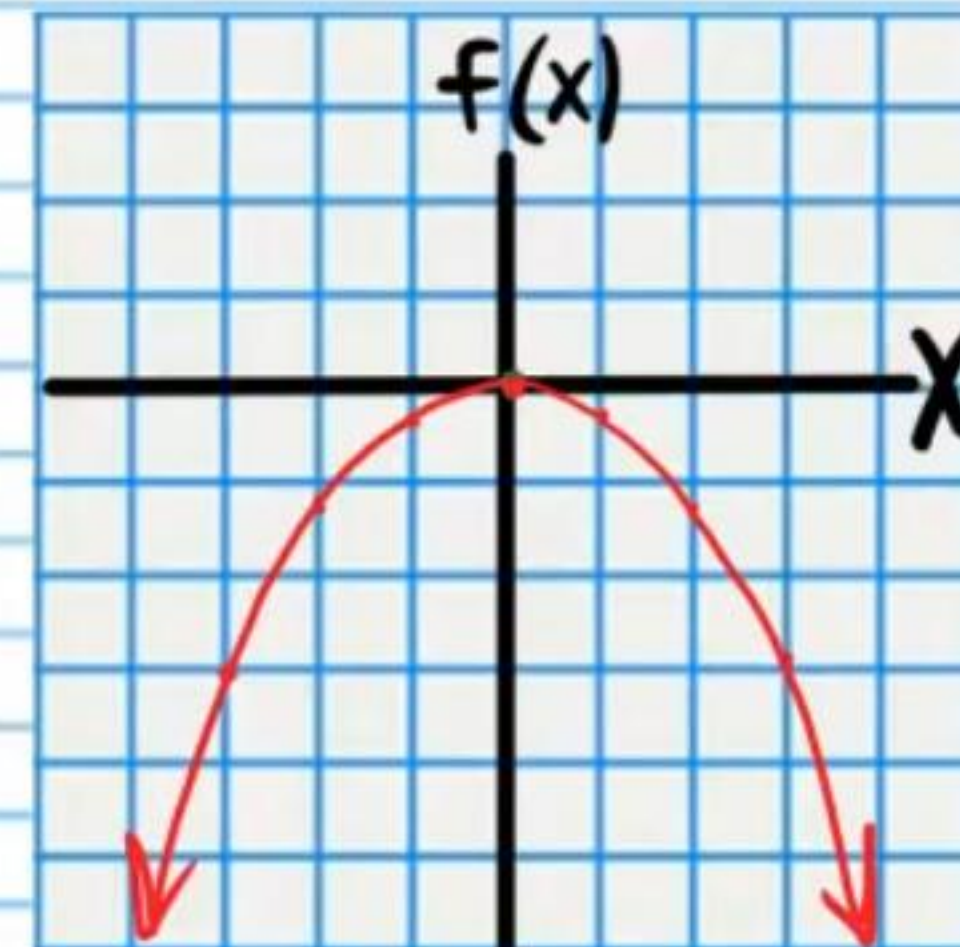
$-3y = x^2$

$y = x^2$

$3y = x^2$

$3(-y) = x^2$

$-3y = x^2$

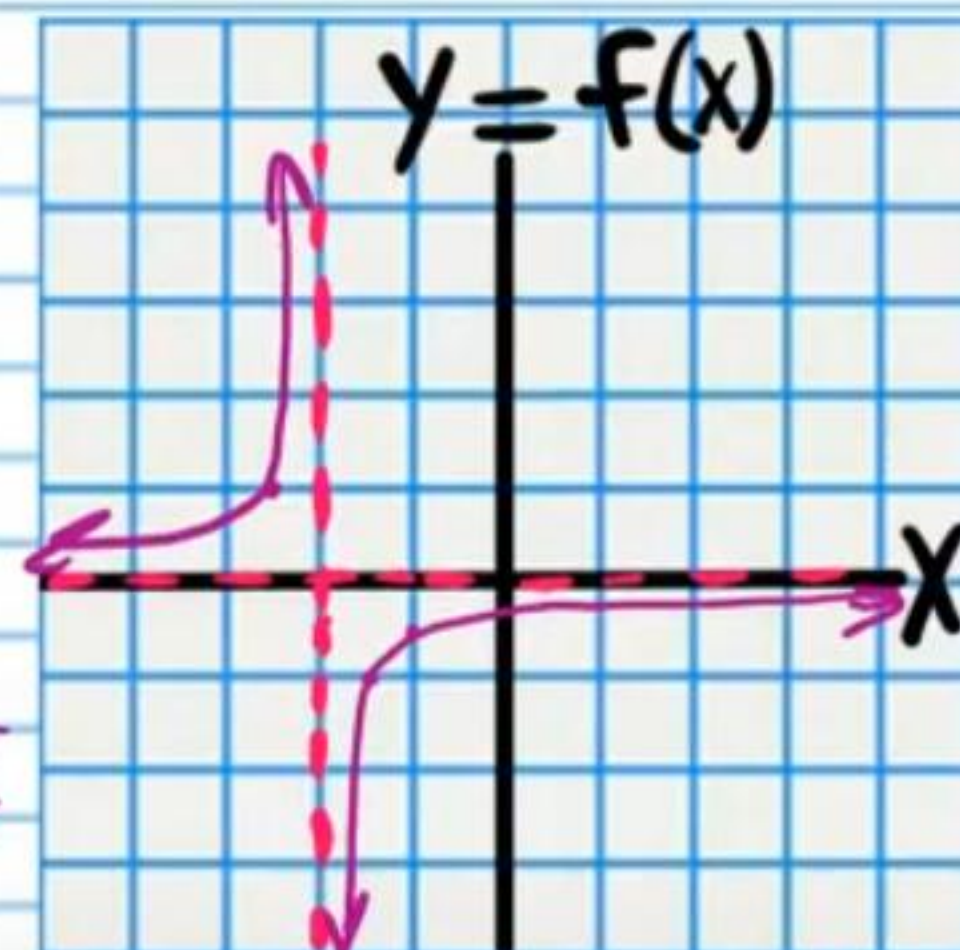


⑤⑪ $f(x) = \frac{1}{-2x-4}$

$y = \frac{1}{x}$

$y = \frac{1}{x-4}$

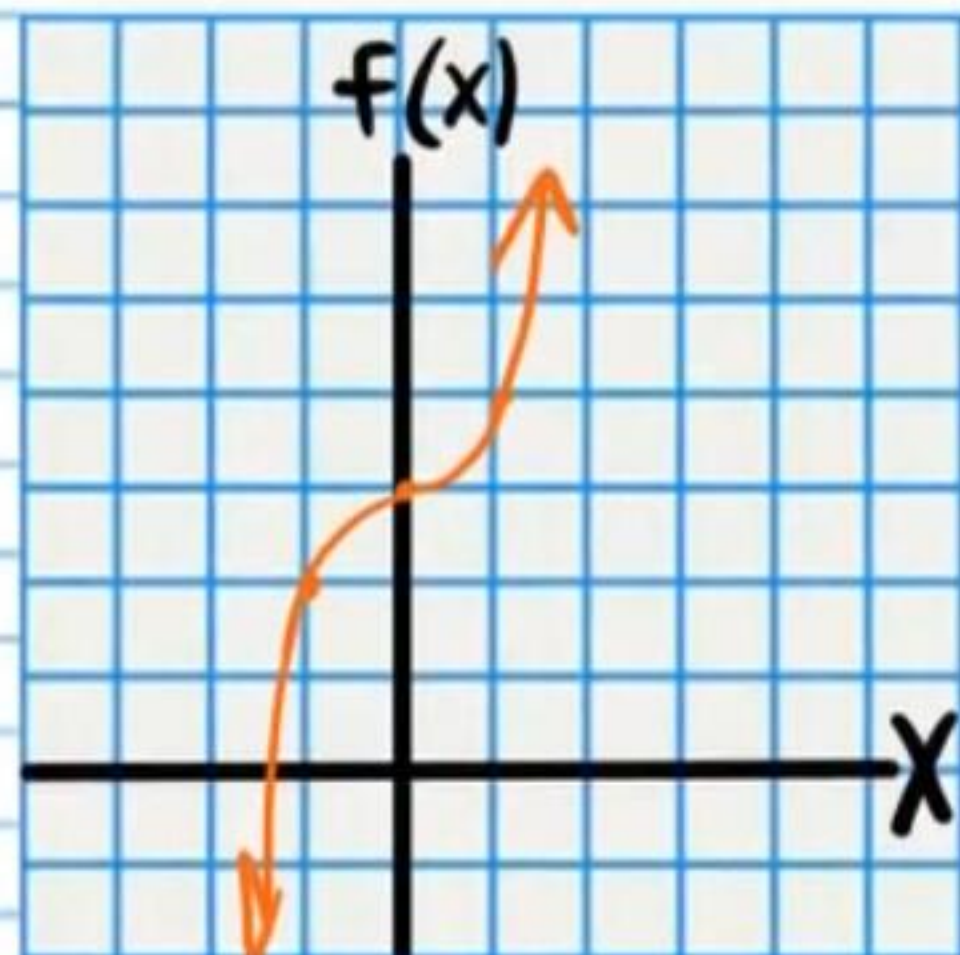
$y = \frac{1}{-2x-4}$



52) $f(x) = x^3 + 3$

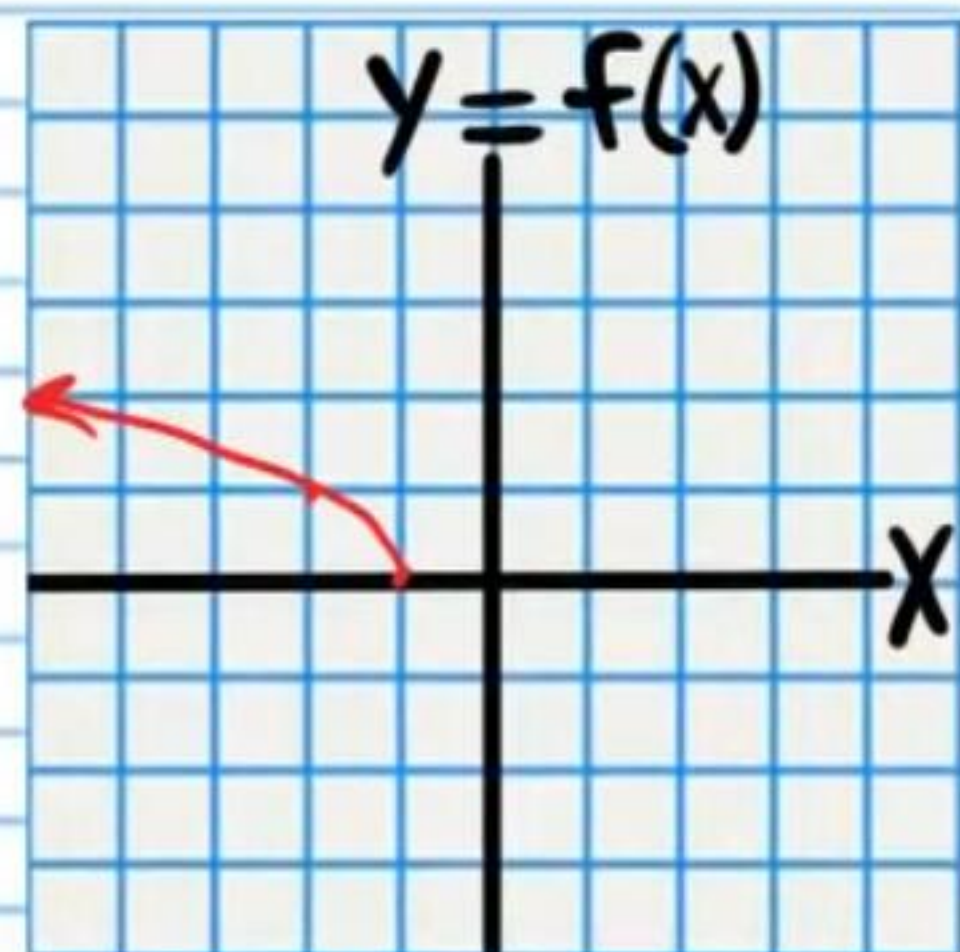
$\downarrow$
 $y = x^3 + 3$
 $y - 3 = x^3$

$y = x^3$
 $y - 3 = x^3$



53) $f(x) = \sqrt{-x-1}$

$y = \sqrt{x}$
 $y = \sqrt{x-1}$
 $y = \sqrt{-x-1}$



54) $f(x) = 2^{-\frac{x-4}{4}}$

$y = 2^x$
 $y = 2^{-\frac{x}{4}}$
 $y = 2^{-\frac{x-4}{4}}$

